

12 Identifying two unknown solutions**[10]**

A student carries out tests on two unknown solutions, A and B. The results are shown in the table.

Solution	Test	Observation	Inference to be completed
A	Flame test	lilac flame	
A	Acidify, then add aqueous silver nitrate	white precipitate	
B	Add aqueous sodium hydroxide	green precipitate; surface slowly turns brown	
B	Acidify, then add aqueous barium nitrate	white precipitate	

(a) Identify the positive ion and negative ion present in solution A. **[2]**

(b) Give the formula of the dissolved salt in solution A. **[1]**

(c) Identify the positive ion and negative ion present in solution B. **[2]**

(d) Give the formula of the dissolved salt in solution B. **[1]**

(e) Write the ionic equation, including state symbols, for the sulfate precipitation test. **[2]**
